YCCE

Vision

"To become the most preferred institution providing innovative, research and value based, professional education for the society at large".

Mission

YCCE is committed to

- Attract best talent and create learning ambience
- Practice Innovative teaching-learning & research
- Integrate Industry-Institute Collaborations
- Nurture students towards holistic development and choicest career

Department

Vision of the Department

To be a well-known center for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary arena by developing problem-solving skills through emerging technologies.

Nagar Yuwak Shikshan Sanstha's

Yeshwantrao Chavan College of Engineering

(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)

Hingna Road, Wanadongri, Nagpur - 441 110

Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu

Department of Computer Science and Engineering (IOT)

23CT1402	Lab: Operating Systems

Name of the Student: Sheikh Wasimuddin	Semester/ Section: 5 A
Roll No: 64	Enrollment Number: 23071044

Sr. No.	COs	POs												PSOs	
		Course Outcomes	1	2	3	4	5	6	7	8	9	10	11	PSO 1	PSO2
1	CO1	Demonstrate the ability to execute Linux process management, memory management, and shell commands to manage system resources efficiently.	3	3	3	-		-	-	-	-	-	3	3	
2	CO2	Develop programs utilizing system calls, thread programming, and page replacement algorithms to simulate and analyze operating system functionalities.	3	3	3	-	-	-	-	-	-	-	3	3	
3	CO3	Design and implement process scheduling, memory allocation, and deadlock detection algorithms to address real-world operating system challenges.	3	3	3	-		-	-	-	-	-	3	3	
		Avg		3	3	-		-	-	-	-	-	3	3	

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Practical No. 1

Aim: Explore Configuration details of Computer System Hardware and Operating System of your personal computer/Laptop with respect to -

a) Processor model
b) RAM size
c) Processor frequency
d) Cache memory size
e) SSD and HDD capacities
f) Operating system (32-bit or 64-bit)
g) Hyper-threading support: Indicate if the processor supports hyper-threading and provide a brief explanation.

Requirement (Hardware/Software) :

A personal computer or laptop with Windows/Linux OS to check hardware configuration, and necessary system tools (Task Manager, Command Prompt, or Settings) to retrieve processor, memory, storage, and OS details.

***Theory**:*

a) **Processor model: -** Identifies the specific type and generation of the CPU, which determines processing capability and features.

b) **RAM size: -** The amount of Random Access Memory available, affecting how many tasks/programs can run simultaneously.

c) **Processor frequency:** The speed at which the CPU executes instructions, measured in GHz; higher values generally mean faster performance.

d) **Cache memory size:** Small, high-speed memory inside the CPU used to store frequently accessed data for faster processing.

e) **SSD and HDD capacities:** Total storage space; SSDs are faster and more efficient, while HDDs offer higher capacity at lower cost.

f) **Operating system (32-bit or 64-bit):** Indicates OS architecture; 64-bit supports more RAM and can handle larger data sizes than 32-bit.

g) **Hyper-threading support:** A CPU technology that allows each physical core to handle two tasks (threads) at once, improving multitasking.

Configuration of My system *(Desktop, Laptop and Mobile):*

A. Mobile(Redmi Note 12 Pro 5G)

a) Processor Model – MediaTek Dimensity 1080, 6 nm, octa-core.
b) RAM Size – 8 GB+8GB expandable.
c) Processor Frequency – Performance cores 2.6 GHz, efficiency cores 2.0 GHz.
d) Cache Memory Size – doesn't have a dedicated cache memory size.
e) SSD and HDD Capacities – 256 GB UFS 2.2 internal storage (no HDD).
f) Operating System (32-bit or 64-bit) – 64-bit Android 14.
g) Hyper-threading Support – Not supported on ARM CPUs (uses multi-core parallelism).

B. Laptop (HP Victus 16-s0095AX)

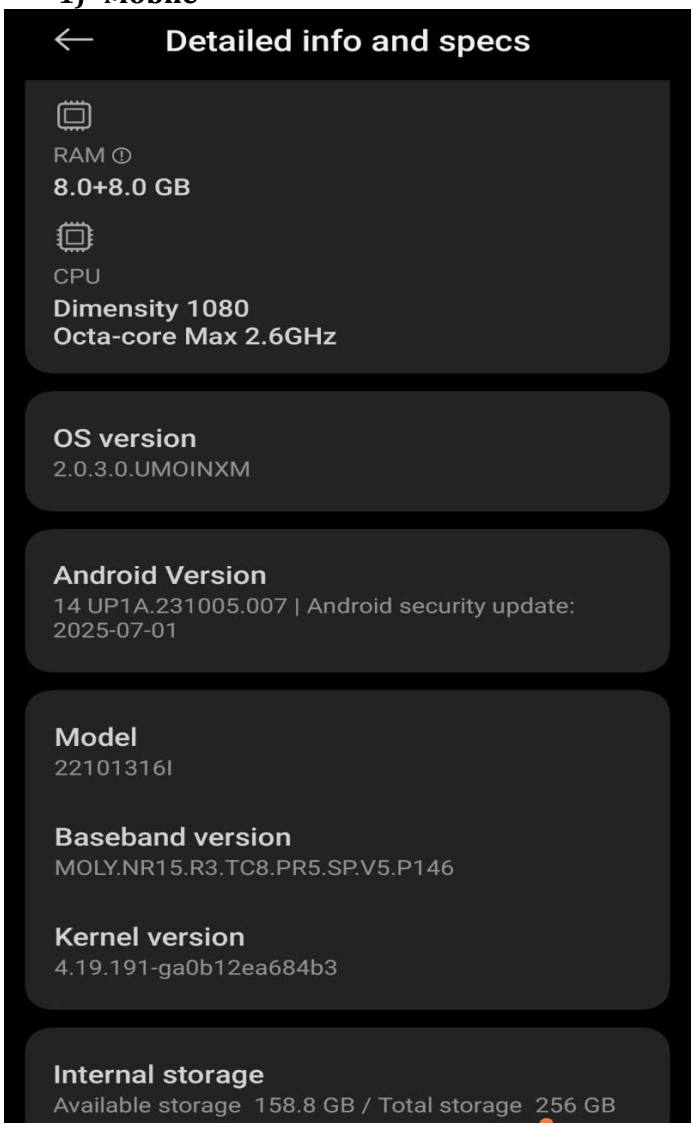
a) Processor Model – AMD Ryzen 7 7840HS (8-core)
b) RAM Size – 16 GB (expandable up to 64 GB at 5600 MHz).
c) Processor Frequency – Max boost clocks up to 5.1 GHz.
d) Cache Memory Size – 16 MB L3 cache.
e) SSD and HDD Capacities – 512 GB SSD , no HDD included.
f) Operating System (32-bit or 64-bit) – Windows 11 Home (64-bit).
g) Hyper-threading Support – uses AMD's Zen multi-threading; 16 threads on 8 cores.

C. Desktop:

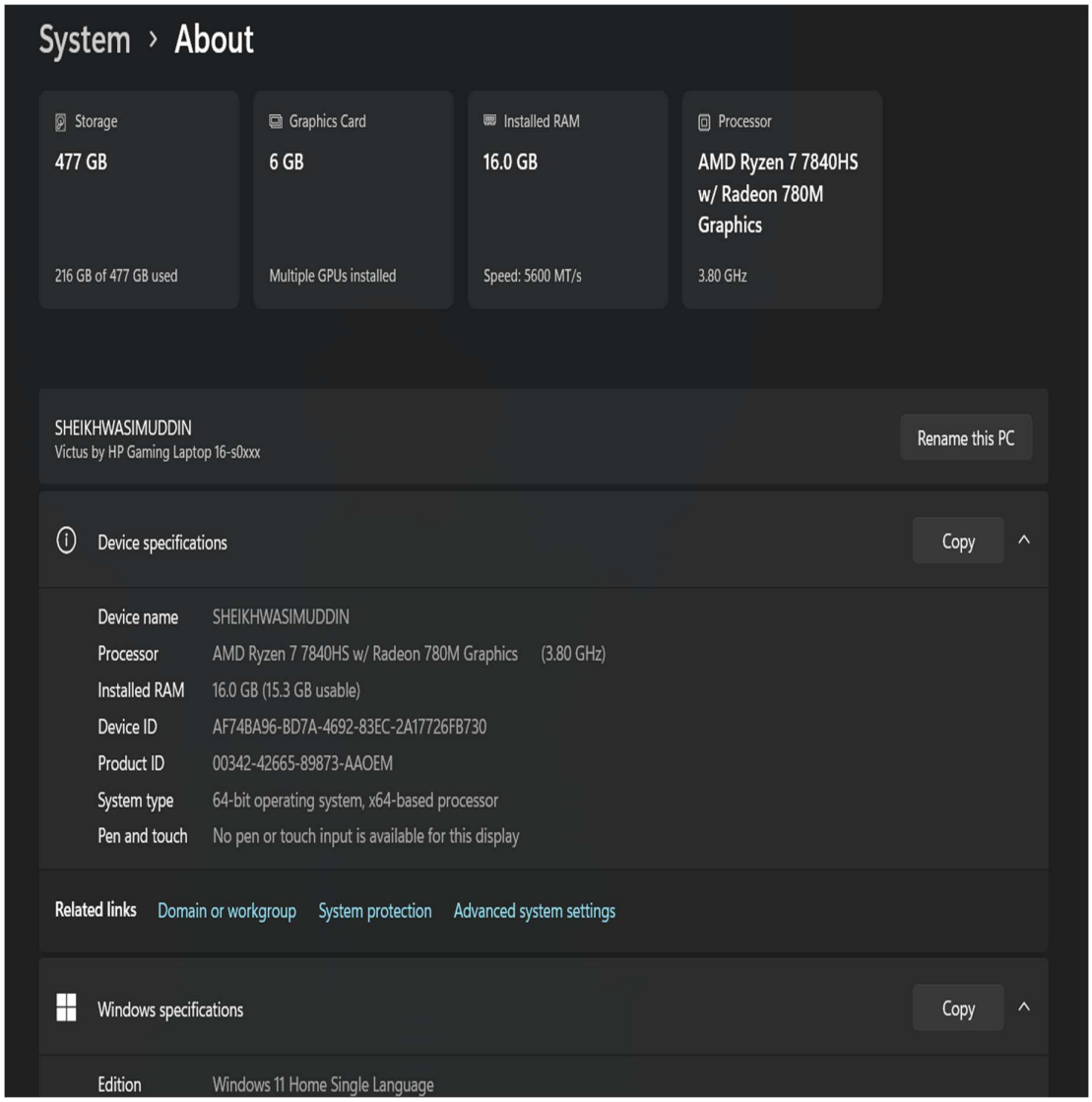
a) Processor Model – AMD Ryzen 7 5700G with Radeon Graphics
b) RAM Size – 16 GB DDR4
c) Processor Frequency – 2.9 GHz base, up to 4.8 GHz turbo
d) Cache Memory Size – L1: 512 KB, L2: 2 MB, L3: 16 MB
e) SSD and HDD Capacities – 1 TB SSD + 2 TB HDD
f) Operating System (32-bit or 64-bit) – Windows 11 Education, 64-bit OS
g) Hyper-threading Support – Supported; enables simultaneous execution of two threads per core

OUTPUT (SCREEN SHOT) :

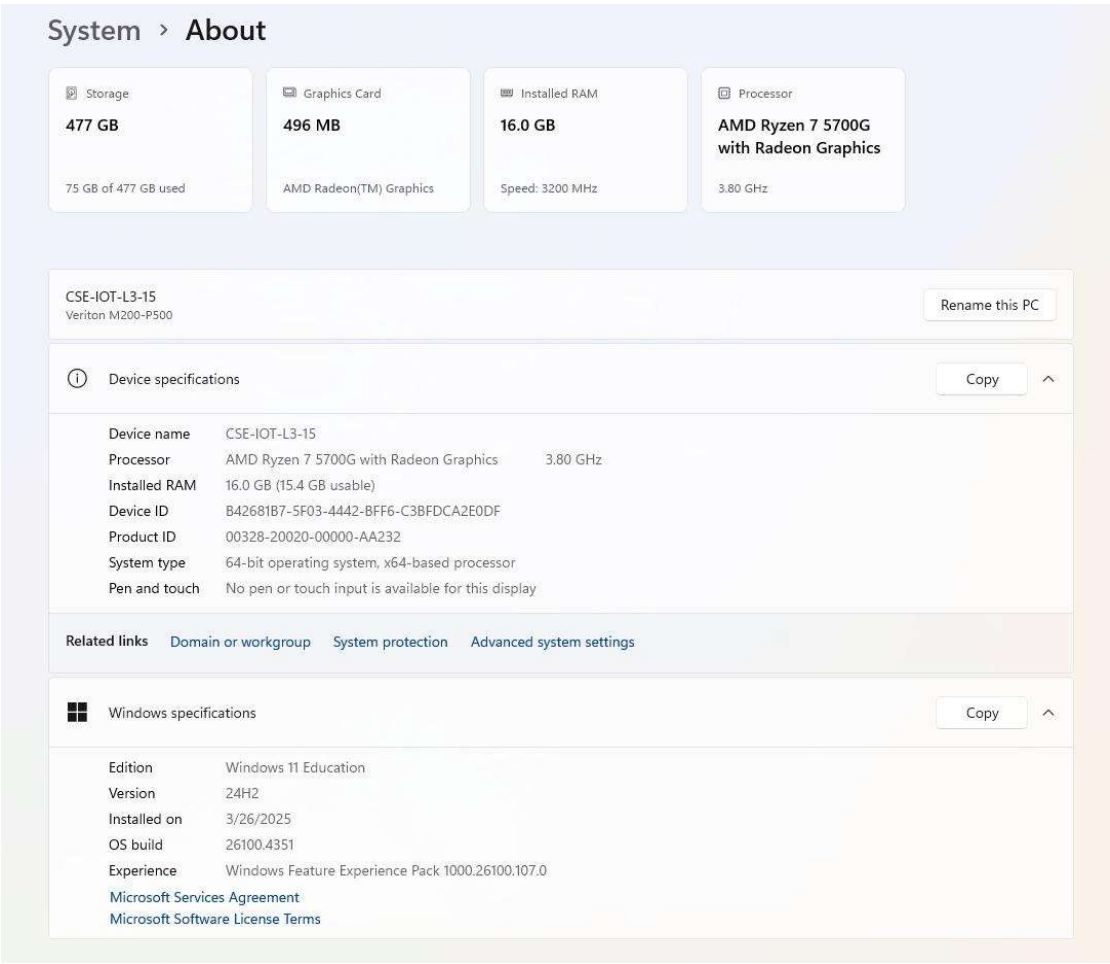
1) **Mobile**



2) Laptop:


System › About
Storage
477 GB
216 GB of 477 GB used
Graphics Card
6 GB
Multiple GPUs installed
Installed RAM
16.0 GB
Speed: 5600 MT/s
Processor
AMD Ryzen 7 7840HS w/ Radeon 780M Graphics
3.80 GHz
SHEIKHWASIMUDDIN
Victus by HP Gaming Laptop 16-s0xxx
Rename this PC
Device specifications
Copy
Device name SHEIKHWASIMUDDIN
Processor AMD Ryzen 7 7840HS w/ Radeon 780M Graphics (3.80 GHz)
Installed RAM 16.0 GB (15.3 GB usable)
Device ID AF74BA96-BD7A-4692-83EC-2A17726FB730
Product ID 00342-42665-89873-AAOEM
System type 64-bit operating system, x64-based processor
Pen and touch No pen or touch input is available for this display
Related links Domain or workgroup System protection Advanced system settings
Windows specifications
Copy
Edition Windows 11 Home Single Language

3) Desktop:



Conclusion: Understanding system configuration helps compare device performance and choose the right one for tasks. Specifications like CPU, RAM, storage, and OS play a key role in efficiency.